

An Analysis of Generative adversarial network

Generative adversarial network

$$\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$$

Generative adversarial network -- Part 1

CycleGAN CycleGAN is an architecture for performing translations between two domains, such as between photos of horses and photos of zebras, or photos of night cities and photos of day cities.

Generative adversarial network -- Part 2

Main theorems for GAN game The original GAN paper proved the following two theorems: Interpretation: For any fixed generator strategy μ_G , the optimal discriminator keeps track of the likelihood ratio between the reference distribution and the generator distribution: $D(x) = \frac{\mu_{\text{ref}}(x)}{\mu_{\text{ref}}(x) + \mu_G(x)}$; $D(x) = \sigma(\ln \frac{\mu_{\text{ref}}(x)}{\mu_G(x)})$ where σ is the logistic function. In particular, if the prior probability for an image x to come from the reference distribution is equal to $\frac{1}{2}$, then $D(x)$ is just the posterior probability that x came from the reference distribution: $D(x) = \Pr(x \text{ came from reference distribution} \mid x)$.

Generative adversarial network -- Part 3

Knight, Will. "5 Big Predictions for Artificial Intelligence in 2017". MIT Technology Review. Retrieved January 5, 2017. Karras, Tero; Laine, Samuli; Aila, Timo (2018). "A Style-Based Generator Architecture for Generative Adversarial Networks". arXiv:1812.04948 [cs.NE]. This Person Does Not Exist – photorealistic images of people who do not exist, generated by StyleGAN This Cat Does Not Exist Archived March 5, 2019, at the Wayback Machine – photorealistic images of cats who do not exist, generated by StyleGAN Wang, Zhengwei; She, Qi; Ward, Tomas E. (2019). "Generative Adversarial Networks in Computer Vision: A Survey and Taxonomy". arXiv:1906.01529 [cs.LG].

Generative adversarial network -- Part 4

Unstable convergence While the GAN game has a unique global equilibrium point when both the generator and discriminator have access to their entire strategy sets, the equilibrium is no longer guaranteed when they have a restricted strategy set. In practice, the generator has access only to measures of form $\mu_Z \circ G_\theta^{-1}$, where G_θ is a function computed by a neural network with parameters θ , and μ_Z is an easily sampled distribution, such as the uniform or normal distribution. Similarly, the discriminator has access only to functions of form D_ζ , a function computed by a neural network with parameters ζ . These restricted strategy sets take up a vanishingly small proportion of their entire strategy sets. Further, even if an equilibrium still exists, it can only be found by searching in the high-dimensional space of all possible neural network functions. The standard strategy of using gradient descent to find the equilibrium often does not work for GAN, and often the game "collapses" into one of several failure modes. To improve the convergence stability, some training strategies start with an easier task, such as generating low-resolution images or simple images (one object with uniform background), and gradually increase the difficulty of the task during training. This essentially translates to applying a curriculum learning scheme.

Generative adversarial network -- Part 5

One of its purposes is to solve the problem of mode collapse (see above). The authors claim "In no experiment did we see evidence of mode collapse for the WGAN algorithm".

Generative adversarial network -- Part 6

GANs with alternative architectures The GAN game is a general framework and can be run with any reasonable parametrization of the generator G and discriminator D . In the original paper, the authors demonstrated it using multilayer perceptron networks and convolutional neural networks. Many alternative architectures have been tried. Deep convolutional GAN (DCGAN): For both generator and discriminator, uses only deep networks consisting entirely of convolution-deconvolution layers, that is, fully convolutional networks. Self-attention GAN (SAGAN): Starts with the DCGAN, then adds residually-connected standard self-attention modules to the generator and discriminator. Variational autoencoder GAN (VAEGAN): Uses a variational autoencoder (VAE) for the generator. Transformer GAN (TransGAN): Uses the pure transformer architecture for both the generator and discriminator, entirely devoid of

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convolution-deconvolution layers. Flow-GAN: Uses flow-based generative model for the generator, allowing efficient computation of the likelihood function.

Generative adversarial network -- Part 7

Evaluation GANs are usually evaluated by Inception score (IS), which measures how varied the generator's outputs are (as classified by an image classifier, usually Inception-v3), or Fréchet inception distance (FID), which measures how similar the generator's outputs are to a reference set (as classified by a learned image featurizer, such as Inception-v3 without its final layer). Many papers that propose new GAN architectures for image generation report how their architectures break the state of the art on FID or IS. Another evaluation method is the Learned Perceptual Image Patch Similarity (LPIPS), which starts with a learned image featurizer $f_\theta : \text{Image} \rightarrow \mathbb{R}^n$, and finetunes it by supervised learning on a set of (x, x') , perceptual difference $\mathcal{D}(x, x')$, where x is an image, x' is a perturbed version of it, and perceptual difference $\mathcal{D}(x, x')$ is how much they differ, as reported by human subjects. The model is finetuned so that it can approximate $\mathcal{D}(x, x') \approx \|f_\theta(x) - f_\theta(x')\|$. This finetuned model is then used to define LPIPS $\mathcal{LPIPS}(x, x') := \|f_\theta(x) - f_\theta(x')\|$. Other evaluation methods are reviewed in.